Nihal Mohan Shettigar

Email: nihalmohan46@gmail.com
LinkedIn | GitHub | Portfolio
Mobile: +91-9980434633

Python engineer building AI-native platforms and enterprise automation systems. Experienced across full stack development, LLM orchestration, async backend architecture, and cloud infrastructure.

Experience

- **LatSpace** — Remote
  Founding Engineer (Part Time) — *Nov 2025 – April 2026*
  - Architected and built core backend services for an AI-native ESG compliance platform using FastAPI, async Python, and MongoDB with Beanie ODM, serving CBAM and CCTS regulatory workflows
  - Designed LangGraph-based AI agent systems with modular node architecture for automated ESG data processing, integrating structured LLM outputs validated through Pydantic v2 schemas
  - Built multi-provider LLM gateway (Bifrost) abstracting OpenAI, Anthropic, and AWS Bedrock for unified model routing and structured response handling across all AI services
  - Developed Next.js 16 frontend with React 19 Server/Client Components, TanStack Query v5 for server state, and Zustand for client state management with strict TypeScript throughout
  - Set up Docker Compose orchestration across 5 microservices, multi-stage Dockerfiles for production, and AWS infrastructure (EC2, S3, IAM) with multi-AZ failover and SOC 2 compliance
  - Enforced engineering quality standards including 66% test coverage as CI gate, Ruff linting, Mypy strict typing, and pre-commit hooks across the platform
- **Cognizant Technology Solutions** — Chennai, India
  Programmer Analyst — *June 2025 – Present*
  - Performed data validation for SAP ECC to S/4HANA conversion, ensuring data integrity across enterprise migration workflows involving large-scale datasets
  - Developed VBS-based automation tooling and Excel-driven reporting pipelines, reducing manual operational effort in enterprise data processing workflows
  - Gained exposure to enterprise-scale systems, structured process workflows, and operational reliability requirements in large organizational environments

Internships

- **National Institute of Technology Karnataka** — Surathkal, Mangalore
  Research Intern — *Jan 2025 – May 2025*
  - Developed a Video Captioning model integrating state-of-the-art computer vision and sequence-to-sequence methods for efficient caption generation from video input
  - Implemented end-to-end training pipeline with frame extraction, feature encoding, and attention-based decoding, contributing to ongoing research in multimodal AI

Projects

- **Sketchify – Sketch to Realistic Face Generator**
  GAN, Deep Learning, Python — *Final Year Project — Oct 2024*
  - Built a GAN-based sketch-to-image generator trained on CelebAMask-HQ dataset, producing realistic face images from raw hand-drawn sketch inputs
  - Developed a web interface for sketch input and real-time inference, with backend processing through the trained generative model

Technical Skills

- **Languages**: Python, TypeScript, JavaScript, SQL
- **Backend**: FastAPI, Async Python, Pydantic v2, REST APIs, JWT Authentication, Beanie ODM
- **Frontend**: Next.js, React, TypeScript, TanStack Query, Zustand
- **AI & Agents**: LangGraph, OpenAI API, Anthropic API, AWS Bedrock, Structured LLM Pipelines
- **Databases**: MongoDB, Motor (async driver)
- **Infrastructure**: Docker, Docker Compose, AWS (EC2, S3, IAM), CI/CD, Multi-stage Builds
- **Dev Tools**: Git, Claude Code, Codex, pytest, Ruff, Mypy, Pre-commit Hooks
- **Enterprise**: SAP ECC, SAP S/4HANA, VBS Automation, Excel Automation

Education

- **NMAM Institute of Technology** — Nitte, Karnataka
  Bachelor of Engineering in Artificial Intelligence and Machine Learning; CGPA: 7.98 — *2021 – 2025*